WHICH TOOL? MATCH WHAT THE QUESTION SAYS

Count arrangements, committees, seats, chips from a batch	Past paper: Counting and hypergeometric
Events per minute / per year, "no more than 2"	Past paper: Poisson, scale λ to the interval
"Which source is more likely to exceed..."	Past paper: Q((t−μ)/σ) for each, compare
Decide 0 or 1 from a noisy received value	Past paper: threshold from f(y|x)P(x) rule
A PDF has an unknown constant	Past paper: area = 1, then E, Var, CDF
Count successes in n independent trials	Binomial(n,p)
Trials until the first success	Geometric(p), memoryless
Waiting time, time to failure, arrivals	Exponential(λ), memoryless
Noise, measurement error, many small effects	Gaussian N(μ,σ²), then standardise
Every value in a range equally likely	Uniform(a,b)
Probability at one exact value (continuous)	Always 0. Use an interval or the CDF

Add or average n measurements	Sum/mean rules; CLT for the shape
"As n grows, does the estimate converge?"	Law of Large Numbers
"Approximately normal for large n"	Central Limit Theorem
"Best guess" for a parameter	Point estimate ($\bar{x}$, s^2). Sample variance divides by n−1
"Plus or minus", "how confident"	Confidence interval $\bar{x} \pm z \cdot \sigma/\sqrt{n}$
"How many samples for ±E"	$n = (z\sigma/E)^2$, round up
Fit a line / plane through data	Linear regression, $w = X^+y$, add a column of 1s for bias
Curved data, choose degree	Polynomial features + BIC (take the minimum)
No closed form, iterative training	Gradient descent, $w \leftarrow w - \eta\nabla L$
Predict a class label (2 classes / K classes)	Sigmoid + BCE / softmax + cross-entropy
Overfitting, weights too large	Ridge: $w = (X^TX + \alpha I)^{-1}X^Ty$

TOPICS

Exam What the 2026 formula sheet and Z-table give you

Use when: Checked against the ECE2191 formula sheet and the Z-table.

Given: vector norm, inner product, cosine similarity
Needs: **Trap:**

Given: $W = (X^TX)^{-1}X^TY$ for multivariate regression, MSE, R^2, BIC, gradient descent update
Needs: Tall X (more samples than features). **Trap:** The fat-matrix form $X^T(XX^T)^{-1}$ is NOT on the sheet.

Given: sigmoid, cross-entropy, softmax, sign, tanh, ReLU, MSE, BCE, NLL
Needs: **Trap:** Hinge loss and the log loss in practice 6 are not listed.

Given: quadratic formula, product/quotient/chain rule, derivatives of x^n, e^x, ln, sin, cos, tan, hyperbolics
Needs: **Trap:** $\sigma' = \sigma(1-\sigma)$ is not listed. Derive it or keep it from this sheet.

NOT given: any probability formula
Needs: No Binomial, Geometric, Poisson, Exponential, Gaussian PDF, CLT, CI, sample size, Bayes, covariance. **Trap:** Put these first in your notes.

NOT given: ridge, precision, recall, F1, C(n+k,k) feature count, one-vs-rest, X^+ for fat matrices
Needs: **Trap:** All on this sheet.

Z-table gives $\Phi(z) = P(Z \le z)$ to 4 decimals, for negative and positive z
Needs: $\Phi(-z) = 1 - \Phi(z)$. $Q(z) = 1 - \Phi(z)$. **Trap:** Earlier practice papers gave a Q-table. The 2026 table is Φ, so subtract from 1 for tails.

Table values to expect: $\Phi(1)=0.8413$, $\Phi(1.5)=0.9332$, $\Phi(1.6)=0.9452$, $\Phi(1.96)=0.9750$, $\Phi(2)=0.9772$, $\Phi(2.8)=0.9974$
Needs: **Trap:** Q(2)=0.0228, Q(1.5)=0.0668, Q(1.6)=0.0548, Q(2.8)=0.0026 with the 4-decimal table.

Row is z to one decimal, column is the second decimal
Needs: $\Phi(1.96)$: row 1.9, column 0.06. **Trap:** Values can differ in the 4th decimal from software.

Exam Counting and hypergeometric

Use when: Seen: practice 4 Q1, practice 2 Q6, practice 3 Q3.

Round table, n people: $(n-1)!$ arrangements (fix one person)
Needs: Rotations identical. **Trap:** Next to spouse among 20: $2 \cdot 18!/19! = 2/19$.

Choose k from n: $C(n,k) = n!/(k!(n-k)!)$
Needs: Order does not matter. **Trap:**

Hypergeometric: P(exactly j faulty in k from N with M faulty) = $C(M,j)C(N-M,k-j)/C(N,k)$
Needs: Draw without replacement. **Trap:** Chips: 30 total, 11 broken, test 5, exactly 1 faulty: C(11,1)C(19,4)/C(30,5).

Sample space of the count of faulty chips among 5 drawn: {0,1,2,3,4,5}
Needs: Limited by how many faulty chips exist. **Trap:** List values, not outcomes.

Groups by category: product of $C(\cdot,\cdot)$ for each category, divided by C(total, group size)
Needs: Clayton and Malaysia, male and female: 4 categories. **Trap:** Check the chosen counts add to the group size.

Joint PMF from random draws (X reds, Y whites among 3 drawn from 2R, 3W, 4B): $P = C(2,x)C(3,y)C(4,3-x-y)/C(9,3)$
Needs: Marginals: sum the joint over the other variable. **Trap:** X and Y are not independent (sample without replacement).

Stuck? Try: Write the total number of equally likely outcomes first, then count the favourable ones.

Exam Poisson counts

Use when: Seen: practice 2 Q1, practice 3 Q2. Not in your weekly notes, but on the papers.

$P(N=k) = e^{-\lambda}\lambda^k / k!$, k=0,1,2,... $E[N] = \mathrm{Var}(N) = \lambda$
Needs: Events independent, constant average rate. **Trap:** λ is for THE INTERVAL ASKED. Scale it first.

Rate scaling: λ = rate × interval length
Needs: Same time unit on both. **Trap:** 600/min over 200 ms: $\lambda = 600 \times (0.2/60) = 2$. 1460/year over one day: $\lambda = 4$.

$P(N=0) = e^{-\lambda}$
Needs: **Trap:** $\lambda=2 \Rightarrow 0.1353$. $\lambda=4 \Rightarrow 0.0183$.

$P(N \le 2) = e^{-\lambda}(1 + \lambda + \lambda^2/2)$
Needs: Add the terms k=0,1,2. **Trap:** $\lambda=2 \Rightarrow 5e^{-2} = 0.677$. $\lambda=4 \Rightarrow 13e^{-4} = 0.238$.

Time between Poisson events is Exponential(λ per unit time)
Needs: **Trap:** Link to the memoryless waiting-time model.

Exam Gaussian tail comparisons and sample means

Use when: Seen: practice 2 Q2, practice 3 Q1, Q6, practice 4 Q7, mid-sem Q8.

$P(X > t) = Q((t-\mu)/\sigma)$

Needs: X Gaussian. **Trap:** Source A (230, 5) vs B (225, 10) above 240: Q(2)=0.0228 vs Q(1.5)=0.0668. B more likely.

Lifetime ≥ 30,000: Sensor 1 (28000, 1000): Q(2)=0.0228. Sensor 2 (24000, 4000): Q(1.5)=0.0668

Needs: Choose the larger probability. **Trap:** Choose Sensor 2.

Sample mean of n Gaussian readings: $\bar{X} \sim N(\mu, \sigma^2/n)$

Needs: Use σ/√n as the SD. **Trap:** Mid-sem Q8: (13.2−12)/(6/√64) = 1.6 ⇒ 1−Φ(1.6) = 0.0548 (Z-table Φ(1.6)=0.9452).

$P(45 < \bar{X} < 52)$, σ=10, n=16: SD = 2.5, z from 0 to 2.8

Needs: Q(0) − Q(2.8) **Trap:** = 0.5 − 0.002555 = 0.4974.

Symmetric interval with probability p: $\mu \pm z\cdot\sigma/\sqrt{n}$ with $Q(z) = (1-p)/2$

Needs: 95% ⇒ z=1.96. 60% ⇒ Q(z)=0.2 ⇒ z≈0.84. **Trap:** 95% of sample means: 45 ± 1.96(2.5) = (40.1, 49.9).

Required n so that $|\bar{X} - \mu| < \sigma/10$ with probability 0.95: $n = (1.96 \times 10)^2$

Needs: **Trap:** = 384.16 ⇒ n = 385. Round up.

Peak height of a Gaussian PDF is $1/(\sigma\sqrt{2\pi})$: bigger peak = smaller σ

Needs: **Trap:** f(0) is a density, not P(A=0). P(A=0) = 0 for both.

Exam PDF with an unknown constant

Use when: Seen: practice 2 Q3, practice 3 Q4, practice 4 Q2.

Total area = 1 fixes the constant

Needs: ∫ f dx = 1 over the support. **Trap:** f = a(4−(x−38)²) on [36,40]: area = a·32/3 ⇒ a = 3/32.

Second condition (e.g. E[X] given) fixes a second constant

Needs: f = a x^b on [0,1]: a = b+1, E = (b+1)/(b+2) = 0.6 ⇒ b = 0.5, a = 1.5. **Trap:**

CDF: F(x) = ∫ from lower limit to x. Write 0 below and 1 above

Needs: Piecewise. **Trap:** $F(x) = x^{1.5}$ on [0,1].

$P(X \geq c) = 1 - F(c)$

Needs: **Trap:** Symmetric PDF about 38 ⇒ P(X ≥ 38) = ½ and E = 38.

$E[X] = \int x\, f$. $E[X^2] = \int x^2\, f$. $Var = E[X^2] - E[X]^2$

Needs: **Trap:** $f = 1.5x^{0.5}$: $E[X^2] = 3/7$, Var = 3/7 − 0.36 = 0.0686.

$Y = aX + b$: $E[Y] = aE[X] + b$, $Var(Y) = a^2Var(X)$

Needs: **Trap:** Y = 2X+1: Var = 4Var(X). °F = 1.8·°C + 32: $Var_F = 3.24\, Var_C$.

Piecewise PDF read from a graph: area = 1 gives a; Y = g(X) gives a PMF by summing probabilities of regions

Needs: **Trap:** P(X = 1) = 0 for continuous X.

Exam Joint distributions: independent? uncorrelated?

Use when: Seen: practice 3 Q5, mid-sem Q3, Q15, practice 4 Q5.

Independence test: P(x,y) = P(x)P(y) for every cell. One failing cell proves dependence

Needs: **Trap:** p(0,1)=p(1,0)=p(2,1)=1/3: P(X=0,Y=0)=0 but P(X=0)P(Y=0) = 1/9. Dependent.

Uncorrelated test: Cov = E[XY] − E[X]E[Y] = 0

Needs: **Trap:** Same example: E[X]=1, E[Y]=2/3, E[XY]=2/3 ⇒ Cov = 0. Uncorrelated but dependent.

X uniform on {−1,0,1}, $Y = X^2$: E[XY] = 0 ⇒ uncorrelated, but Y is a function of X

Needs: **Trap:** Zero covariance never proves independence.

Conditional PDF: $f(y|x) = f(x,y)/f_X(x)$, $f_X(x) = \int f(x,y)dy$

Needs: Fix x, integrate y over its limits at that x. **Trap:** f = 2 on 0<y<x<1: f_X = 2x, f(y|x) = 1/x, so P(Y < x/2 | X=x) = (x/2)/x = ½.

E[aX+bY] = aE[X] + bE[Y]. Cov(F, aF+b) = a Var(F)

Needs: Linearity of expectation needs no independence. **Trap:** Practice 4 Q5: prove it from the integral definition.

Exam Bayes, total probability, event trees

Use when: Seen: practice 2 Q6c.

$P(A) = \Sigma\, P(A|B_i)P(B_i)$ (total probability)

Needs: B_i partition the space. **Trap:** P(not faulty) = 0.8·(5/30) + 0.6·(25/30) = 0.6333.

$P(B_i|A) = P(A|B_i)P(B_i)/P(A)$

Needs: **Trap:** P(new | not faulty) = 0.1333/0.6333 = 0.2105.

Draw the tree and multiply along branches

Needs: Leaves sum to 1. **Trap:**

Exam Binary detection with Gaussian noise

Use when: Seen: practice 4 Q3, practice 5 mixed.

Decide "0" if $f_Y(y|X=-1)P(X=-1) > f_Y(y|X=1)P(X=1)$

Needs: Y = X + N, N ~ N(0,σ²). **Trap:**

Threshold: $T = (\sigma^2/2)\ln(P_0/P_1)$

Needs: Gaussian likelihoods with equal σ, means −1 and +1. **Trap:** $P_0 = 2P_1$, $\sigma^2 = 1/16$ ⇒ T = (1/32)ln 2 = 0.0217.

$P(\text{error}) = P_0\, Q((T+1)/\sigma) + P_1\, Q((1-T)/\sigma)$

Needs: σ = 0.25 here. **Trap:** Q(4.09) and Q(3.91) are about 2×10^{-5} and 4.6×10^{-5}. Use the Q table.

Equal priors ⇒ T = 0

Needs: **Trap:** Unequal priors move the threshold toward the less likely symbol.

Stuck? Try: Take logs of both sides, collect y on one side, and you get a straight threshold on y.

Exam Concept traps from the mid-semester test

Use when: Multiple choice. Know why each wrong option is wrong.

Bursty errors: $S = \Sigma X_i$ with $P(X_i=1)=0.01$. E[S] = 5 still holds

Needs: Linearity needs no independence. **Trap:** S is NOT Binomial and Var(S) is NOT np(1−p) without independence.

Geometric: 5 failures already; P(more than 3 more attempts) = 0.8^3

Needs: Memoryless. **Trap:** Do not use 0.8^8 or 0.8^5.

CDF flat on (2.0, 2.5) ⇒ probability of that interval is 0

Needs: **Trap:** It does not mean uniform or discrete.

CLT with n=30 and $P(X_i=1)=0.001$: $P(\bar{X}=0) = 0.999^{30} = 0.97$, so Gaussian approximation is poor

Needs: Rule n ≥ 30 is not a theorem. Very skewed parents need far larger n. **Trap:**

LLN with a drifting offset D_i: not identically distributed, average drift need not vanish

Needs: IID is required. **Trap:** More samples do not remove a systematic offset.

$\hat{X}_A = X_1$ and $\hat{X}_B = \bar{X}$ are both unbiased. $\hat{X}_B$ has SE $\sigma/\sqrt{n}$

Needs: **Trap:**

95% CI (−0.08, 0.12) inside tolerance ±0.20 supports "mean bias is within tolerance"

Needs: **Trap:** Not "95% of sensors" and not "95% of future errors".

σ unknown, replace by S: interval follows Student t, not exact with z

Needs: Gaussian data. **Trap:**

Halve the margin of error ⇒ quadruple n

Needs: **Trap:**

Exponential with mean 200 h: λ = 0.005, Var = $1/\lambda^2$ = 40,000, SD = 200

Needs: Mean equals SD. **Trap:** "Most fail near 200 h" is FALSE. The PDF is largest at 0 and decays. The bathtub (useful-life) remark does not answer the question.

Exam

Regression computations and properties (AI practice exam)

Use when: Seen: practice 6 Q1, Q3 to Q9, mock Q3 and Q6.

Residual $e_i = y_i - \hat{y}_i$. MSE $= (1/m)\Sigma e_i^2$

Needs: **Trap:** Least-squares fit with bias: $\Sigma e_i = 0$ and $\Sigma x_i\, e_i = 0$ and $\Sigma \hat{y}_i\, e_i = 0$.

Recover lost residuals using $\Sigma e=0$ and $\Sigma xe=0$ (two equations, two unknowns)

Needs: Fitted by least squares with a bias. **Trap:** Then $y_i = \hat{y}_i + e_i$.

Rescale the features: least squares gives the same predictions, ridge does not

Needs: Same α. **Trap:** Ridge penalises $\|w\|^2$, which changes when units change.

$x'=2x-1$, $y'=10y+5$: $w' = 5w$, R^2 unchanged, MSE $\times$ 100

Needs: Affine changes of x and y. **Trap:** $b' = 10b + 5w + 5$.

Fit y on x and x on y: $w_1 w_2 = R^2$, lines coincide only if $R^2=1$

Needs: S_xy, S_xx, S_yy. w_1 = S_xy/S_xx, w_2 = S_xy/S_yy. **Trap:** Both lines pass through $(\bar{x}, \bar{y})$. SSE = S_yy − S_xy²/S_xx.

$R^2 = 1 - \text{SSE/SST}$. Higher is better on training data

Needs: SST = $\Sigma(y-\bar{y})^2$. **Trap:** R^2 on training data always improves with more features. Use validation or BIC.

Fewer samples than parameters: $w\dagger = X^T(XX^T)^{-1}y$ (minimum norm)

Needs: Under-determined. **Trap:** x=(1,2), y=5: $w\dagger$ = (1,2). Prediction at (2,1) = 4, but another zero-error w=(5,0) predicts 10. Zero training error says little.

Features like cos x, sin x, x^2: still linear regression

Needs: Linear in the parameters. **Trap:** Design matrix columns (1, cos x, sin x). With four equally spaced phases X^TX is diagonal.

Outliers pull the MSE line toward themselves

Needs: **Trap:** Spot large residuals, check the data, remove or use a robust loss, then refit.

Polynomial features of degree $\le$ k from n inputs: C(n+k,k)

Needs: n=3: k=2 ⇒ 10, k=3 ⇒ 20. **Trap:** 20 parameters with 12 samples is under-determined. Training MSE ≈ 0. Do not trust it.

Exam

BIC and model selection numbers

Use when: Seen: practice 6 Q2, Q10.

BIC $= d \ln m + m \ln(\text{MSE})$

Needs: Natural log. Lower is better. **Trap:** m=50: A (d=3, 0.40) = −34.08. B (d=6, 0.33) = −31.96. A is selected.

Complexity term $d \ln m$ grows with degree. Fit term $m \ln$ MSE falls

Needs: The sum has a minimum. **Trap:**

B beats A only if $\text{MSE}_B < 0.316$

Needs: Solve $6 \ln m + m \ln M < 3 \ln m + m \ln 0.40$. **Trap:**

With $m = 5000$ the fit term dominates: B is selected

Needs: **Trap:** More data makes extra parameters easier to justify.

Exam

Gradient descent questions

Use when: Seen: practice 5 mixed, practice 6 Q12 to Q16.

Training loop: initialise w, repeat $w \leftarrow w - \eta \nabla L$ until a stopping rule

Needs: $\eta > 0$. **Trap:** State the stopping rule: gradient small, loss change small, or max iterations.

$f = \frac{1}{2}x^2 - 4x + 12$: $\nabla f = x - 4$. One step to the minimum needs $\eta = 1$

Needs: Quadratic with curvature 1. **Trap:** General: $f = (c/2)x^2$-type ⇒ best one-step $\eta = 1/\text{curvature}$.

$f(w) = (w-2)^2+1$: $w^{(t)} - 2 = (1-2\eta)^t(w^{(0)}-2)$

Needs: **Trap:** $w^{(0)}=5$, $w^{(1)}=3.8$ ⇒ $\eta=0.2$. Then 3.08, 2.648. $\eta=1$ oscillates 5, −1 forever.

Convergence from every start: $|1 - \eta\lambda| < 1$ for every curvature λ

Needs: $f = \frac{1}{2}(\omega_1^2 + 9\omega_2^2)$ ⇒ $0 < \eta < 2/9$. **Trap:** The steepest direction limits the step size.

Learning rate choice: start moderate, watch the loss curve, decay when it plateaus

Needs: Too big: oscillate or diverge. Too small: slow. **Trap:** A faster, still smooth curve after a change means η was raised.

Gradient of $f(x_1,x_2) = (x_1-3)^3\sigma(x_2)$: $\partial/\partial x_1 = 3(x_1-3)^2\sigma(x_2)$, $\partial/\partial x_2 = (x_1-3)^3\sigma(x_2)(1-\sigma(x_2))$

Needs: $\sigma' = \sigma(1-\sigma)$. **Trap:** At (−1,1) ≈ (35.09, −12.58).

Exam

Classification questions

Use when: Seen: practice 5 mixed, practice 6 Q17 to Q22.

Why not linear regression for classes: unbounded output, wrong loss, integer labels imply order

Needs: **Trap:** State at least two reasons.

K classes with only logistic regression: K one-vs-rest models

Needs: 3 classes ⇒ 3 models. **Trap:** Pick the class with the largest probability.

Softmax uses exp: positive outputs, sum to 1. $z_i/\Sigma z$ fails with negatives or a zero sum

Needs: **Trap:**

Remove all activations from an MLP: it collapses to a single linear model

Needs: Composition of linear maps is linear. **Trap:**

BCE per sample: $-y \ln \hat{y} - (1-y)\ln(1-\hat{y})$. Gradient $(\hat{y}-y)(x,1)$

Needs: $y \in \{0,1\}$. **Trap:** The sample with the largest $|\hat{y}-y|$ has the largest gradient. Confident wrong predictions dominate.

Hinge loss $\max(0, 1-yf)$: for $y=-1$ minimised for any $f \le -1$ (greatest such value is −1). Log loss $\ln(1+e^{-yf})$: no finite minimiser

Needs: **Trap:**

Linear classifier $\text{sign}(w \cdot x + b)$: boundary $w_1x_1 + w_2x_2 + b = 0$, w is normal to it

Needs: Side with $w \cdot x + b \ge 0$ is class +1. **Trap:** w=(2,−3), b=1: line $x_2 = (2/3)x_1 + 1/3$.

Metrics at a threshold: accuracy (TP+TN)/N, precision TP/(TP+FP), recall TP/(TP+FN)

Needs: Raise the threshold: precision up, recall down. **Trap:** If a missed fault is dangerous, choose the lower threshold (higher recall).

ReLU network: $f = \text{ReLU}(x) - 2\text{ReLU}(x-1) + \text{ReLU}(x-2)$ is a triangle with peak 1 at x=1, zero outside [0,2]

Needs: Piecewise linear, one kink per hidden unit. **Trap:**

Quick version: past-paper question types only. The full sheet has the week-by-week reference.